RHCA White Test - Solutions

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1 Machine 1 Solutions

1.1 Network Configuration

First, configure the network settings:

```
# Configure hostname
hostnamectl hostname client.example.com
```

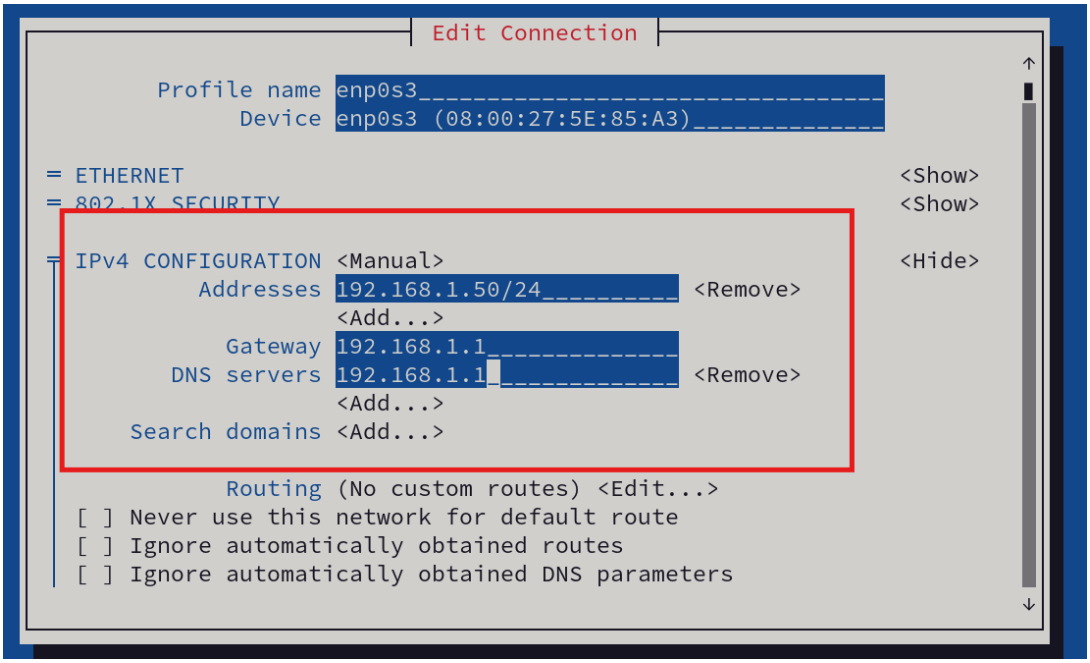


Figure 1: nmtui Edit Connection Interface

1. Set multi-user target as default

```
systemctl set-default multi-user.target
reboot
```

2. Find lines containing "seismic"

```
grep "seismic" /usr/share/dict/words > /root/wordlist
```

3. Create user David with password aging

```
useradd David
echo "tekup" | passwd --stdin David
chage -m 7 -M 28 -W 5 David
```

4. Create groups with shared GID

```
groupadd -g 5000 linuxadm
groupadd -g 5000 -o dba
usermod -aG linuxadm David
```

5. Modify group name and GID

```
groupmod -n sysadm -g 6000 linuxadm
usermod -g sysadm David
```

6. Create archive directory

```
mkdir /archive
chmod 770 /archive
systemctl list-unit-files --type=target > /archive/targets
```

7. Create user melinda

```
groupadd -g 3200 melinda_group
useradd -u 4525 -g 3200 melinda
echo "redhat" | passwd --stdin melinda
```

8. Configure default permissions for melinda

```
# Add to melinda's .bashrc or .bash_profile
su - melinda
echo "umask 0012" >> ~/.bashrc
# Or modify /etc/profile for system-wide effect
```

9. Copy archive with specific permissions

```
cp -r /archive /root/archive2
chown -R david:sysadm /root/archive2
chmod g+s /root/archive2
```

10. Create a services file with specific permissions

```
mkdir -p /srv/admin
systemctl list-units --type=service > /srv/admin/services

# Set ownership: melinda as owner, sysadm as group
chown melinda:sysadm /srv/admin/services

# Set permissions:
# - melinda (owner) can read and write (delete)
# - sysadm group members can read
# - others have no access
chmod 640 /srv/admin/services

# Verify permissions
ls -l /srv/admin/services
# Should show: -rw-r----- 1 melinda sysadm ... services

# Test access (optional verification)
# su - melinda -c "cat /srv/admin/services"  # Should work
# su - david -c "cat /srv/admin/services"    # Should work (david is
    in sysadm)
```

11. Configure YUM repositories

```
# Create /root/BaseOS.repo
cat > /root/BaseOS.repo << EOF
[BaseOS]
name=BaseOS Repository
baseurl=http://content.example.com/rhel9.0/x86_64/dvd/BaseOS
enabled=1
gpgcheck=0
EOF

# Create /root/AppStream.repo
cat > /root/AppStream.repo << EOF
[AppStream]
name=AppStream Repository
baseurl=http://content.example.com/rhel9.0/x86_64/dvd/AppStream
enabled=1
gpgcheck=0
EOF
```

12. Create backup archive

```
cd /root
tar -cjf backup.tar.bz2 /usr/local
```

13. Configure AutoFS

```
# Install autofs
yum install -y autofs

# Create the AutoFS configuration file
cat > /root/AutoFS << 'EOF'
-----------------------------------------------------
First file to modify : /etc/auto.master
Content of the first file : /home/guests /etc/auto.misc

Second file to modify : /etc/auto.misc
Content of the second file : * -rw classroom.example.com:/home/guests
   /&

The Command you execute after all of this : systemctl enable --now
   autofs
-----------------------------------------------------
EOF
```

14. Create and run a PDF converter container

```
# Create user athena
useradd athena
echo "trootent" | passwd --stdin athena

# Install container tools
yum install -y podman

# Create directories
mkdir -p /data/input /data/output
chown -R athena:athena /data
chmod -R 777 /data

# Switch to athena user
su - athena

# Download files
wget https://raw.githubusercontent.com/sachinyadav3496/Text-To-PDF/
   master/pdf_converter.py
wget https://raw.githubusercontent.com/sachinyadav3496/Text-To-PDF/
   master/Dockerfile

# Build container
podman build -t pdf .

# Run container
podman run -d --name pdfconverter \
  -v /data/input:/data/input:Z \
  -v /data/output:/data/output:Z \
  pdf

# Create test file
echo "Test content for PDF conversion" > /data/input/file.txt
```

15. Configure systemd service for the PDF converter container

```
# As athena user
su - athena

# Create systemd user directory
mkdir -p ~/.config/systemd/user

# Generate systemd service
podman generate systemd --name pdfconverter --files

# Move service file
mv container-pdfconverter.service ~/.config/systemd/user/

# Enable and start service
systemctl --user daemon-reload
systemctl --user enable container-pdfconverter.service
systemctl --user start container-pdfconverter.service

# Enable user linger (as root)
exit
loginctl enable-linger athena
```

16. Configure HTTP web server

```
  # Install httpd
  yum install -y httpd

  # Configure to listen on port 87
  sed -i 's/Listen 80/Listen 87/' /etc/httpd/conf/httpd.conf

  # Allow port 87 in SELinux
  semanage port -a -t http_port_t -p tcp 87

 # Configure firewall
 firewall-cmd --permanent --add-port=87/tcp
 firewall-cmd --reload

 # Start and enable httpd
 systemctl enable --now httpd
```

17. Write file search script

```
cat > /root/find_david_files.sh << 'EOF'
#!/bin/bash
find / -user david -size +10k -size -10M 2>/dev/null > /tmp/find1
EOF

chmod +x /root/find_david_files.sh
./find_david_files.sh
```

18. Configure NTP client

```
# Install chrony
yum install -y chrony

# Configure chrony
echo "server africa.pool.ntp.org iburst" >> /etc/chrony.conf

# Restart and enable chrony
systemctl enable --now chronyd
```

2 Machine 2 Solutions

2.1 Network Configuration

Configure the network settings for Machine 2:

```
# Configure hostname
hostnamectl set-hostname server.example.com
```

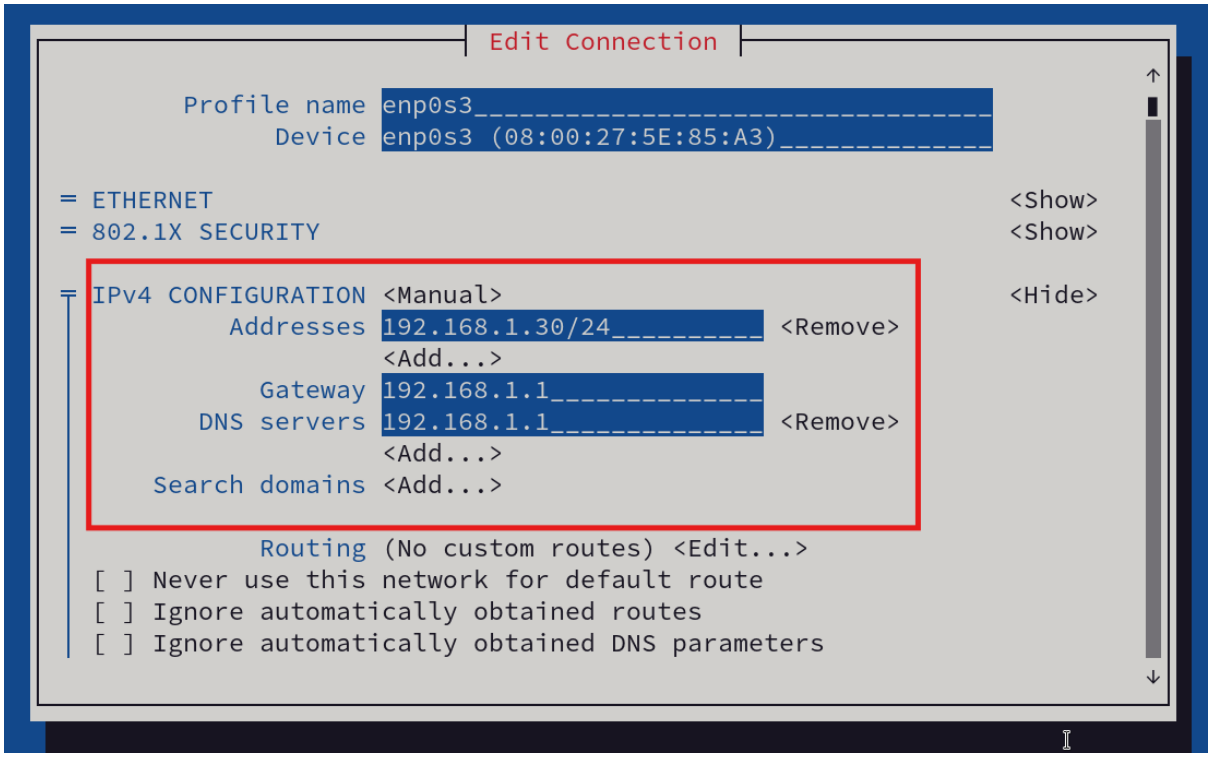


Figure 2: nmtui Edit Connection Interface

1. Reset root password

```
# Boot procedure:
# 1. At GRUB menu, press 'e' to edit
# 2. Find line starting with 'linux'
# 3. Add 'init=/bin/bash' to the end
# 4. Press Ctrl+X to boot
# 5. At emergency prompt:
mount -o remount,rw /
passwd root
# Enter: trootent
touch /.autorelabel

 exec /usr/lib/systemd/systemd
```

2. Configure password policy

```
# Edit /etc/security/pwquality.conf
echo "minlen = 7" >> /etc/security/pwquality.conf
echo "dcredit = -1" >> /etc/security/pwquality.conf
echo "ucredit = -1" >> /etc/security/pwquality.conf
echo "lcredit = -1" >> /etc/security/pwquality.conf
```

3. Write a backup script for small files

```
cat > /root/backup-smallfiles.sh << 'EOF'
#!/bin/bash
mkdir -p /root/backup
find /usr -type f -size -2M -exec cp {} /root/backup/ \; 2>/dev/null
EOF

chmod +x /root/backup-smallfiles.sh
```

4. Schedule the etc backup script

```
# Create backup script
cat > /root/backup-etc.sh << 'EOF'
#!/bin/bash
tar -czf /root/etc-backup.tar.gz /etc
EOF

chmod +x /root/backup-etc.sh

# Add to crontab (11 PM every day except Sunday)
crontab -e
# Add this line:
# 0 23 * * 1-6 /root/backup-etc.sh
echo "0 23 * * 1-6 /root/backup-etc.sh" | crontab -
```

5. Create swap partition

```
# Create 2GB swap partition (assuming /dev/sdb1 is available)
fdisk /dev/sdb
# Create new partition of 2GB, type 82 (Linux swap)

# Format as swap
mkswap /dev/sdb1

# Add to /etc/fstab
echo "/dev/sdb1 swap swap defaults 0 0" >> /etc/fstab

# Enable swap
swapon /dev/sdb1
```

6. Create volume group and logical volumes

```
# Create physical volume
pvcreate /dev/sdb

# Create volume group with 16MB PE size
vgcreate -s 16M vgfs /dev/sdb

# Create logical volumes
lvcreate -l 5 -n ext4vol vgfs
lvcreate -l 5 -n xfsvol vgfs

# Format with filesystems
mkfs.ext4 /dev/vgfs/ext4vol
mkfs.xfs /dev/vgfs/xfsvol

# Create mount points
mkdir /ext4vol /xfsvol

# Add to /etc/fstab
echo "/dev/vgfs/ext4vol /ext4vol ext4 defaults 0 0" >> /etc/fstab
echo "/dev/vgfs/xfsvol /xfsvol xfs defaults 0 0" >> /etc/fstab

# Mount filesystems
mount -a

# Verify
df -h /ext4vol /xfsvol
mount
lsblk
```

7. Extend logical volume

```
# Extend logical volume to 160M (within 157M-175M range)
lvextend -l 5 -r  /dev/vgfs/xfsvol

# Extend XFS filesystem if you forgate the -r in the lvextend
xfs_growfs /xfsvol

# Verify
df -h /xfsvol
```

8. Set tuned profile

```
# Install tuned if not present
yum install -y tuned

# Get recommended profile
tuned-adm recommend

# Set the recommended profile
tuned-adm profile $(tuned-adm recommend)

# Verify
tuned-adm active
```